TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	ZnO Nanomembrane Pyroelectric Pixels for Mid-Infrared (3-5 μm) Sensing			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical/Electronic		Electrical/Electronic	
	Medium Category	Small Category	Medium Category	Small Category
	Electrical and Electronic Components	Sensor components	Semiconductor Devices and Systems	Sensor element
Concept of Proposed Technology	**Core concept and functions** We propose a compact mid-infrared (mid-IR, 3–5 μm) detector platform based on ZnO nanomembrane pyroelectric pixels. The device is a thermally isolated pyroelectric pixel: an engineered mid-IR absorber converts incident radiation into a temperature transient in an ultra-low-heat-capacity ZnO nanomembrane, and the resulting ΔT is converted into a charge signal by the membrane's pyroelectric response. The readout is performed using a charge-sensitive front-end (transimpedance or charge amplifier), enabling direct conversion from absorbed mid-IR power to an electrical output suitable for imaging and spectroscopy modules. **What is new** • Giant-pyroelectric ZnO nanomembranes as a practical transducer: enhanced pyroelectric			

	response in released ultrathin membranes combined with very low heat capacity enables high responsivity and fast response. • Absorber-engineered mid-IR detection with CMOS-compatible processing: the mid-IR spectral response is set by the absorber/cavity design while ZnO provides robust room-temperature transduction; compatible with microfabrication and packaging workflows relevant to SMEs. • A single pixel architecture that can operate at room temperature for practical sensing and can be transitioned to cryogenic calorimetric operation to approach single-photon readiness. • Exploratory option: sliding-ferroelectric 2D stacks (e.g., 3R MoS_2) offer tunable polarization gain; we will assess them as a follow-on enhancement after the ZnO platform is validated. **Proposed device architecture (pixel-level)** • Mid-IR absorber and thermal isolation: a suspended membrane or micro-bridge with engineered thermal conductance (G) to set the time constant. • ZnO nanomembrane pyroelectric layer with top/bottom electrodes to collect pyroelectric charge; membrane release and passivation for low drift; optional biasing for operating-point tuning. • Low-noise charge readout with bandwidth matched to the thermal time constant; optional on-chip multiplexing for arrays. **Primary uses (3-5 μm focus)** • Compact spectroscopy and chemical sensing (e.g., process monitoring, gas or vapor identification using mid-IR signatures). • Thermal signature imaging in the 3-5 μm window for industrial inspection and low-light imaging. • Cryogenic single-photon-ready mid-IR calorimetry for emerging quantum sensing and ultra-low-flux measurements. This proposal builds on PI Shi's demonstrated expertise in enhanced pyroelectricity in ultrathin membranes (**Nature 607, 480–485 (2022), from the PI Jian Shi group**) [1] and translates it into a practical ZnO nanomembrane pixel platform for mid-IR sensing, while keeping sliding-ferroelectric stacks as an exploratory future direction for tunability.
Rationale for Joint R&D Project	**Need for joint R&D** Mid-IR (3–5 μm) sensors are high-value components for spectroscopy, industrial monitoring, and defense/security imaging, but compact detectors with high sensitivity remain difficult to realize without costly cooling or complex integration. ZnO nanomembrane pyroelectric pixels offer a practical route: ultralow heat capacity in released membranes combined with strong pyroelectric transduction and CMOS-compatible processing. However, bridging from a laboratory pixel to an engineering-relevant demonstrator requires packaging, readout integration, module-level constraints, and reliability practices that are best developed jointly

	with an industrial partner. **Why collaboration with a Korean SME is essential** • Define product-relevant specifications for mid-IR modules (optics, filters, interfaces, allowable drift, acceptable noise, and environmental constraints). • Co-develop packaging approaches (vacuum packaging, window materials, thermal isolation structures, and mechanical robustness). • Guide array-readout integration choices (discrete readout and multiplexing strategy). • Perform validation and reliability testing protocols aligned with commercialization needs (temperature cycling, shock/vibration, humidity, and long-duration drift). RPI will contribute the materials/device physics, pixel and array design, and characterization infrastructure; the SME partner could contribute module engineering and translation know-how. The outcome is a demonstrator positioned for follow-on product development under a jointly defined roadmap.
Key Words	1. ZnO nanomembrane 2. pyroelectricity 3. mid-IR (3–5 μm) sensing 4. cryogenic calorimetry 5. sliding ferroelectric (exploratory)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	**Level (3)**	**End of the Project**	**Level (5)**
R&D Objectives	**R&D Objectives** • Demonstrate a room-temperature, single mid-IR (3–5 μm) ZnO nanomembrane pyroelectric pixel with quantified responsivity, NEP, time constant, drift, and stability. • Develop pixel design rules that link absorber efficiency, thermal isolation, and polarization transduction to detector sensitivity and bandwidth. • Design and fabricate a small-format mid-IR array (8x8 to 32x32) using the validated pixel architecture and demonstrate array operation (uniformity, cross-talk, and readout). • Demonstrate cryogenic operation and establish single-photon readiness. • Deliver a packaging and module-interface concept (mechanical, thermal, optical) suitable for SME-led productization. Start TRL (3): laboratory-scale verification of basic device function and performance. End TRL (5): prototype production and performance evaluation of a small-array demonstrator			

	with room-temperature mid-IR operation and cryogenic single-photon-readiness dataset.
R&D Target Specifications	**R&D Target Specifications** • Wavelength band: 3-5 μm (mid-IR). • Single-pixel NEP at 300 K: < 10 nW/√Hz (goal), with a clear noise budget and path to improvement. • Specific detectivity (D*): > 10^6 Jones (baseline) for the single pixel under standard assumptions; stretch goal > 10^7 Jones with improved NEP and absorber coupling. • Pixel area: <= 100 x 100 μm^2 active region; pixel pitch consistent with small arrays. • Absorber efficiency in 3-5 μm band: > 60% (goal) with stable spectral response. • Array size: 8x8 minimum; stretch goal up to 32x32 depending on yield and readout integration. • Cryogenic single-photon readiness.
R&D Plan	**Overall approach** The project is organized into two phases aligned to the TIPA structure: Phase 1 (Jul–Dec 2026) is a pre-study combining single-pixel feasibility experiments with a full Phase-2 plan (market/IP/technical milestones); Phase 2 (Jan 2027–Jun 2029) optimizes single-pixel performance, builds small arrays, and adds cryogenic single-photon-readiness demonstration. Across both phases, the central theme is engineering the thermal–electrical transduction chain: mid-IR absorption → temperature transient → pyroelectric charge → low-noise readout. **Phase 1 (Jul-Dec 2026): materials functions, planning** • ZnO nanomembrane preparation: fabricate/release ultrathin ZnO nanomembranes with stable electrodes and passivation; verify pyroelectric response, drift, and stability versus temperature and bias. • Initial pixel microfabrication: build suspended membrane or micro-bridge pixels with controlled thermal conductance; integrate electrodes and mid-IR absorber. • Market and use-case definition (SME-led): identify the top two product concepts and deliver spec sheets (3–5 μm band, sensitivity/NEP target range, response time/frame rate, operating mode, size/power, packaging, and electrical/readout interfaces). • IP and publication plan (OSAT/TIPA framework): prepare a short background vs. foreground IP summary and an initial invention disclosure/publication plan consistent with program policies; finalize the detailed Phase-2 execution plan and risk register. Phase 2 (Jan 2027–Jun 2029): optimized single pixel + small-array mid-IR demonstration + cryogenic single-photon readiness • Pixel microfabrication: optimize ZnO nanomembrane pixel metrics (thermal isolation, electrode

	design, leakage control, absorber coupling) and improve process repeatability/yield in an array-compatible layout. • Readout and metrology: implement charge-sensitive readout and calibrated mid-IR illumination; extract responsivity, NEP, drift, and 1/f noise; produce a complete noise budget. • Small-array design and fabrication: scale pixel layout to 8x8 (minimum) and, where feasible, 16x16-32x32; develop interconnect and multiplexing strategy compatible with module integration. • Array characterization at 300 K: measure non-uniformity, cross-talk, stability, and frame-rate operation; demonstrate mid-IR scene sensing or spectroscopic signal acquisition. • Cryogenic calorimetric operation: operate optimized pixels (and a subset of array pixels, where appropriate) at cryogenic temperature; use attenuated pulsed mid-IR excitation to collect pulse-height histograms, dark-event rates, timing, and energy resolution (single-photon readiness metrics). • Packaging and validation with SME: implement a small-array package concept (window, thermal design, electrical feedthroughs) and perform reliability screening (temperature cycling, shock/vibration, long-duration drift). **Key milestones (high level)** • M1: Phase 1 materials fabrication and demonstrating initial functions, establishment of full-scale R&D plans (Month 6). • M2: Phase 2 single pixel performance dataset complete (Month 10). • M3: First 8x8 array fabricated and functional at 300 K (Month 12-15). • M4: Array performance demonstrated with stability and calibrated non-uniformity metrics (Month 18-24). • M5: Cryogenic calorimetric dataset demonstrating single-photon readiness metrics (Month 24-36). **Risk management** • Risk: insufficient pyroelectric charge responsivity. Mitigation: optimize thermal isolation, increase absorber efficiency, and use electrode/readout optimization to maximize dP/dT; iterate stack thickness and electrode geometry. • Risk: excess 1/f noise or drift at room temperature. Mitigation: materials quality control, passivation, differential readout, and chopping/lock-in strategies; design for stable operating point. • Risk: array yield and cross-talk. Mitigation: conservative pixel pitch in first arrays, robust interconnect layout, and staged scaling (8x8 to larger formats only after yield is established).

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**RPI (Research Institute) - expected roles** • Lead materials selection, preparation, and characterization of ZnO nanomembranes (growth/release, electrode integration, and pyroelectric characterization). • Design and fabricate single pixels and small arrays (suspended structures, electrodes, absorbers) using RPI microfabrication facilities. • Develop the detector physics model (thermal circuit + polarization transduction) and use it to guide design optimization. • Implement low-noise charge readout and calibrated mid-IR test infrastructure; deliver full performance datasets and noise budgets. • Conduct cryogenic measurements in Phase 2 and quantify single-photon readiness metrics (energy resolution, dark-event rate, timing). • Coordinate technical reviews, documentation, and data packages suitable for translation and SME adoption.
Expected R&R of Korean SME Partner	**Korean SME partner - expected roles** • Co-define product-relevant specifications and application targets for mid-IR modules (spectroscopy, sensing, imaging). • Guide packaging and module integration (optical window/filter selection, mechanical housing, thermal design, electrical interfaces). • Advise on array readout integration strategy and system constraints (power, size, manufacturability). • Advise on Phase 1 market analysis and commercialization planning and coordinate IP discussions.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected outcomes and impact** Technological outcomes: (i) a validated room-temperature mid-IR ZnO nanomembrane pyroelectric single pixel with quantified NEP, responsivity, and stability; (ii) a functioning small-format mid-IR array (8x8 minimum, with stretch to 32x32) demonstrating uniform operation and cross-talk control; (iii) a cryogenic calorimetric dataset establishing single-photon readiness metrics and identifying the dominant noise mechanisms and mitigation steps. Commercialization potential: The platform enables compact mid-IR detector modules suitable for spectroscopy and chemical sensing, with a clear path toward packaged arrays. The Korean SME partner will be positioned to lead follow-on product development (module packaging and qualification) using the data packages and prototypes generated here. Broader impact: Success creates a new sensor modality at the intersection of 2D materials and IR

	detection, strengthening the Semiconductor/Display strategic field with a differentiated mid-IR component technology. Applications include process monitoring, environmental sensing, safety and security, and emerging quantum/ultra-low-flux mid-IR measurements. **Key deliverables** • Phase 1 (Dec 2026): materials demonstration, materials functionality, experimental plans for phase 2. • Phase 2 (Jun 2029): single-pixel prototype + full characterization report and noise budget + small-array prototype + demonstration report (uniformity, cross-talk, frame rate) and cryogenic single-photon-readiness report. • Packaging/module interface concept and SME-oriented transfer package (design files, test protocols, performance baselines).

6. References

(1) Jiang J et al, "Giant Pyroelectricity in Nanomembranes," Nature 607, 480-485 (2022).

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization